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EEEE3054

Renewable Generation Technologies

Solar Cells



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INTERMITTENCY ISSUES

INTERMITTENCY ISSUES

Introduction

- Renewable energy sources provide clean, sustainable electricity but suffer from intermittency and variability due to weather and diurnal cycles.
- These intermittency and variability create several issues to the grid:
  - **Intermittency**: Power output depends on solar irradiance.
  - **Mismatch between demand and generation**: PV often generates maximum power at midday, while demand peaks in the evening.
  - **Grid stability issues**: Sudden fluctuations affect frequency, voltage, and power quality.

Power Imbalance

- In an AC power system, power balance is met when

$$P_{gen}(t) = P_{load}(t) + P_{loss}(t)$$

$P_{gen}(t)$ = total real power generated
$P_{load}(t)$ = total real power consumed
$P_{loss}(t)$ = losses in transmission/distribution

- There is power imbalance if

$$\Delta P(t) = P_{gen}(t) - P_{load}(t) - P_{loss}(t) \neq 0$$

- This imbalance ΔP causes acceleration or deceleration of synchronous machine rotors (or equivalently a mismatch in inverter control setpoints).

Power Imbalance

- Given the swing equation

$$2H\frac{d\omega}{dt} = P_m - P_e - D\Delta\omega$$

$$RoCoF = \frac{df}{dt} \approx \frac{\Delta P}{2H_{sys}S_{base}}$$

where

$RoCoF$ = rate of change of frequency
H_{sys}= system inertia constant (s),
ω = rotor electrical angular speed (rad/s),
P_m = mechanical input power,
P_e = electrical output power,
D = damping factor.
S_{base} = apparent power (MVA)

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Power Imbalance

- If $P_{gen} > P_{load}$: extra energy accelerates rotors → frequency rises.
- If $P_{gen} < P_{load}$: rotors slow down → frequency falls.
- The large inertia constant H_{sys} of synchronous machines resists fast changes.
- With high PV penetration (low inertia), H_{sys} is small, so for the same imbalance ΔP, the $RoCoF$ is much higher.
- In short, the grid with high penetration of renewable energy is sensitive to frequency deviation.

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Example

Suppose:

- System base = 1000 MW,
- Inertia constant H_{sys} = 3 s (low-inertia system),
- Sudden imbalance: ΔP = −100 MW

$$\frac{df}{dt} \approx \frac{\Delta P}{2HS_{base}} = \frac{-100}{2 \times 3 \times 1000} = -0.01607\ Hz/s$$

- So, frequency drops at ~0.017 Hz per second.
- In a traditional system with H_{sys} = 6, $RoCoF$ would be only ~0.008 Hz/s. which is much slower.

Ramp Rate Control

- PV power P_{PV} output fluctuates rapidly, creating "ramps" or quick changes in power.
- These rapid changes in generation (ramping) burdens the grid and forces conventional generators to ramp up and down quickly to meet the changing demand. The net power P_{net} is

$$P_{net}(t) = P_{load}(t) - P_{PV}(t)$$

- Power ramp rate required from dispatchable units is:

$$R(t) = \frac{dP_{net}}{dt} = \frac{dP_{load}}{dt} - \frac{dP_{PV}}{dt}$$

- If $\frac{dP_{PV}}{dt}$ is large negative (PV drops quickly), dispatchable units must provide $-\frac{dP_{PV}}{dt}$ MW/min to compensate the drop.
- These power fluctuations lead to grid instability, voltage fluctuations, and frequency deviations, which can harm sensitive equipment.

INTERMITTENCY ISSUES

Example

- Assume large PV ramp down late afternoon, $\frac{dP_{PV}}{dt}$ = −50 MW/10 min.
- Then dispatchable units must ramp +5 MW/min.
- However, a thermal plant has typical maximum ramp rate of 2 MW/min.
- There is a mismatch, and a need for storage.

Voltage Rise and Reverse Power Flow

- When distributed PV exports power back toward the substation, voltage at a customer point can increase.
- Using a simple single-line model, the approximate complex voltage drop ΔV along a line is:

$$\Delta V \approx \frac{RP + XQ}{V}$$

  where R, X are line resistance and reactance, P, Q are real and reactive power flowing (positive when exported toward the substation), and V is nominal voltage.
- If PV exports real power $P > 0$ back into the feeder, the sign of ΔV can be positive, i.e., voltage rise.

Example

Compute the voltage rise for the following power network:

- R = 0.2 Ω, X = 0.8 Ω (short radial feeder numbers)
- PV injects P = 50 kW = 0.05 MW, Q = 10 kVAr = 0.01 MVAr
- Nominal voltage V_s = 400 V (line-to-line for low voltage network)

$$\Delta V \approx \frac{RP + XQ}{V} = \frac{(0.2 \times 50000) \times (0.8 \times 10000)}{400} = 45\ V$$

So the local voltage magnitude would increase by about 45 V (≈ 11.25% of 400 V).



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ENERGY STORAGE SYSTEM

Role of Energy Storage System

To ensure reliability, stability, and quality of utility grid, energy storage systems (ESS) are essential.

- **Energy Balancing:** Stores excess energy when generation > demand, releases it when demand > generation.
- **Grid Stability:** Provides frequency and voltage regulation to reduce fluctuations.
- **Peak Shaving & Load Shifting:** Store excess PV energy during the day and release during peak demand.
- **Backup Power:** Ensures continuous supply during outages or low generation.
- **Maximizing Renewable Penetration:** Enables higher use of renewables without compromising reliability.

Performance Metrics

- Energy storage system has following technical requirement:
  - Round-trip efficiency
  - Response time
  - Lifetime and cycling
  - Sizing

Round-Trip Efficiency

- Losses occur due to internal resistance, conversion devices, and heat dissipation.
- Round-trip efficiency is the ratio of the energy retrieved from the storage system during discharge to the energy put into it during charging.

$$\eta = \frac{E_{discharge}}{E_{charge}} \times 100\%$$

- Higher efficiency means less energy wasted.
- Among energy storage options, compressed-air energy storage (CAES) has the lowest reported efficiency, and Li-ion batteries have the highest.
- For energy storage coupled with photovoltaics, efficiencies of less than 75% are unlikely to be cost-effective.

Response Time

- Response time is the time taken for the storage system to respond and deliver power when required.
- Some applications (like grid frequency regulation) demand very fast response, while others (like seasonal storage) can tolerate slower response.
- Examples
  - Supercapacitors: Very fast, good for frequency regulation
  - Lithium-ion batteries: Fast, suitable for load shifting and grid balancing
  - Pumped hydro storage: Slower start-up, suited for large-scale load leveling.

Lifetime and Cycling

- The expected operational life of the ESS, usually measured in years or number of charge-discharge cycles before performance degrades below a usable level (often 70–80% of original capacity).
  - Calendar life: Ageing over time even without cycling.
  - Cycle life: How many times the ESS can be charged and discharged.
- Examples
  - Lithium-ion batteries: ~3,000–10,000 cycles (10–15 years).
  - Lead-acid batteries: ~500–1,500 cycles (3–5 years).
  - Flow batteries (e.g., Vanadium Redox): ~10,000+ cycles (15–20 years).
  - Pumped hydro: 30–60 years lifetime.

Sizing

- Sizing is the process of determining the required energy capacity (kWh or MWh) and power rating (kW or MW) of the ESS to meet the intended application.
- **Power rating (kW/MW)** determines the instantaneous power the system can deliver.
- **Energy capacity (kWh/MWh)** dictates the duration it can sustain that power (duration of storage).

Types of Energy Storage System

- Mechanical Storage
  - Pumped Hydro Energy Storage (PHES)
  - Compressed Air Energy Storage (CAES)
  - Flywheels
- Electrochemical Storage (Batteries)
  - Lead-Acid Batteries
  - Lithium-Ion Batteries
  - Flow Batteries (Vanadium Redox, Zinc-Bromine)
  - Sodium-Sulfur (NaS) Batteries
- Electrical Storage
  - Supercapacitors
  - Superconducting Magnetic Energy Storage (SMES)
- Thermal Storage
  - Molten Salt
  - Phase Change Materials (PCMs)

Types of Energy Storage System

Storage Type	Examples	Timescale	RTE Efficiency (%)	Energy Density	Applications
Electrochemical	Lithium-ion, Lead-acid, Flow batteries	Short (min–days)	70–95	High (Li-ion), Low (Lead-acid)	Scalable, good for PV
Mechanical	PHES, CAES, Flywheels	Medium (hours–days)	60–85	Very high (PHS), moderate (Flywheel)	Grid-scale, large footprint
Thermal	Molten salts, Phase change materials	Medium (hours–days)	50–90	Moderate	Better for solar thermal, not PV
Chemical	Hydrogen, Synthetic fuels	Long (days–months)	30–50	Very high (as fuel)	Expensive, Good for seasonal storage
Electrical	Supercapacitors	Very short (sec–min)	90–98	Very low	Best for fast response, not bulk storage



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BATTERY ENERGY STORAGE SYSTEM

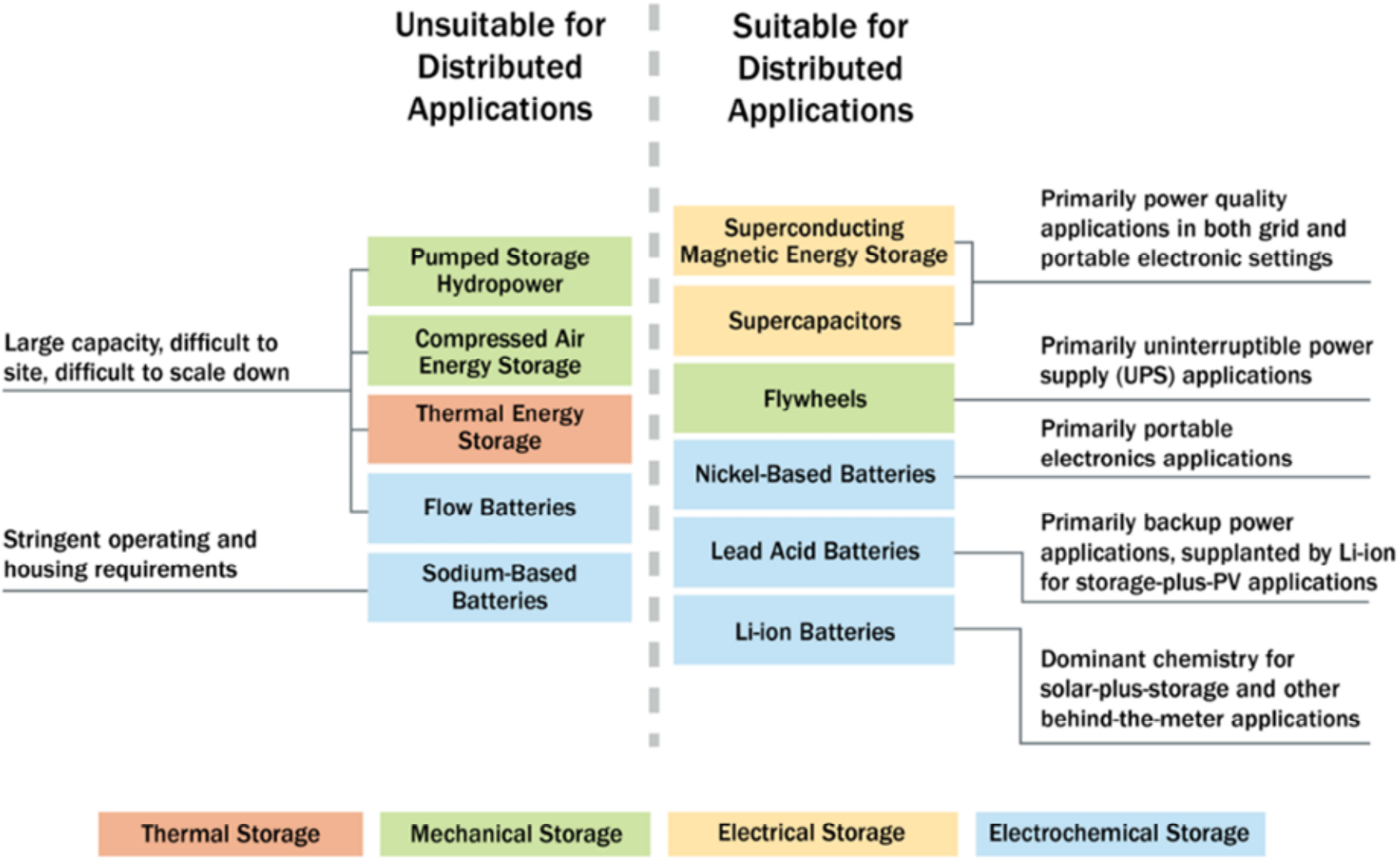
BATTERY ENERGY STORAGE SYSTEM

Why BESS?

- Battery storage is considered the most suitable form of energy storage for PV systems compared to other ESS because it offers high efficiency, scalability, and flexibility in deployment.
- Unlike mechanical ESS (pumped hydro or compressed air), which are location-dependent and suited mainly for large-scale applications, batteries can be installed directly alongside PV arrays in residential, commercial, or utility-scale systems.
- They operate on DC, which matches the output of PV panels, avoiding additional conversion losses and simplifying integration.
- Batteries also respond almost instantaneously to fluctuations in solar generation, making them highly effective for smoothing power output during rapid changes in irradiance.
- Furthermore, declining costs especially in lithium-ion technology have made BESS increasingly affordable and widely adopted.

BATTERY ENERGY STORAGE SYSTEM

Why BESS?



BATTERY ENERGY STORAGE SYSTEM

BESS Control



Battery Controller

- Prevent under/overcharging
- Balance multiple stacks
- Regulate state of charge (SoC) and monitor temperature.



Energy Storage System Controller

- Transient stability
  - Voltage control
  - Frequency control
- Island operation
- Renewable smoothing

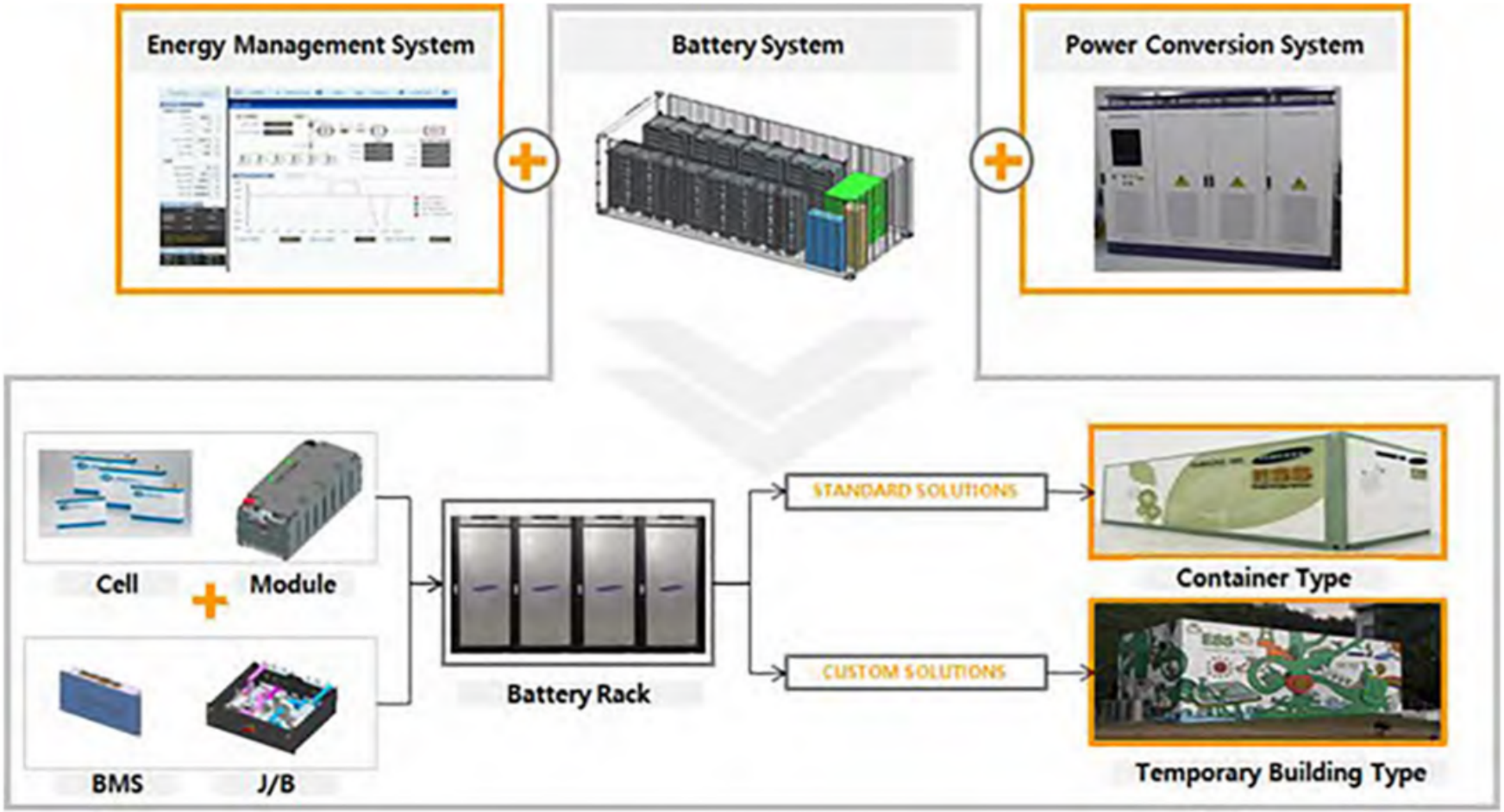


Grid Controller

- Demand response
- Peak shaving
- Time shifting
- Ancillary services.

BATTERY ENERGY STORAGE SYSTEM

BESS Block Diagram



BMS = battery management system, J/B = Junction box.

Source: Korea Battery Industry Association 2017 "Energy storage system technology and business model".

BESS Block Diagram

- **The battery system** consists of the battery pack, which connects multiple cells to appropriate voltage and capacity. T
- **The battery management (system BMS)** protects the cells from harmful operation, in terms of voltage, temperature, and current, to achieve reliable and safe operation, and balances varying cell states-of-charge (SOCs) within a serial connection.
- **The battery thermal management system (B-TMS)** controls the temperature of the cells according to their specifications in terms of absolute values and temperature gradients within the pack.

BESS Block Diagram

- The components required for the reliable operation of the overall system are system control and monitoring, the energy management system (EMS), and system thermal management.
- System control and monitoring is general (IT) monitoring, which is partly combined into the overall supervisory control and data acquisition (SCADA) system but may also include fire and distribution.
- System thermal management controls all functions related to the heating, ventilation, and air-conditioning of the containment system.

BESS Block Diagram

- The power electronics can be grouped into the conversion unit, which converts the power flow between the grid and the battery, and the required control and monitoring components—voltage sensing units and thermal management of power electronics components (fan cooling).

BATTERY ENERGY STORAGE SYSTEM

Types of BESS

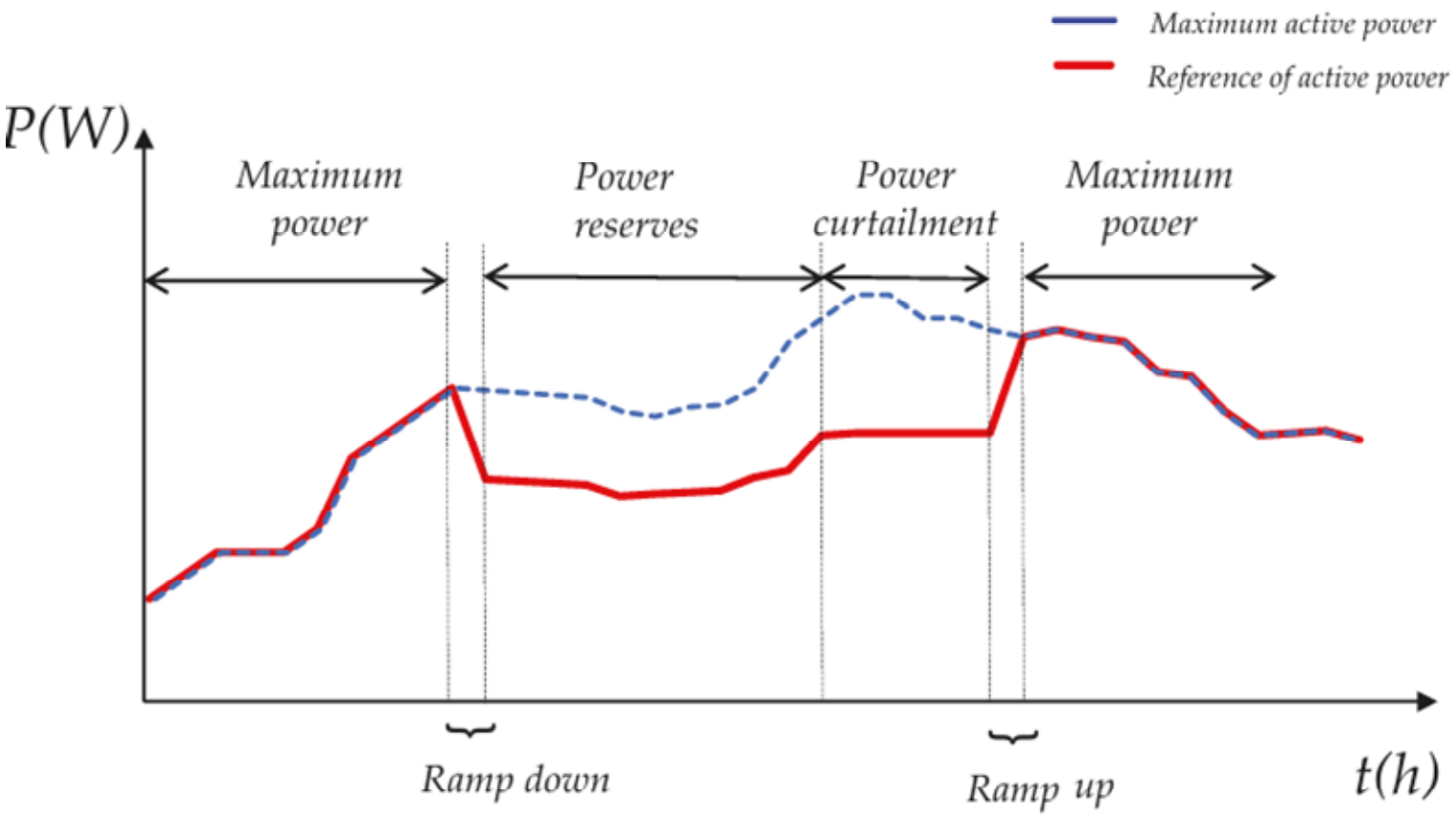
charge and discharge
(**round trip**)

	Energy density (kW/kg)	Round Trip Efficiency (%)	Life Span (years)	Eco-friendliness
Li-ion	1st (150–250)	1st 95	1st (10–15)	Eco-friendliness
NaS	2nd (125–150)	2nd (75–85)	3rd (10–15)	X
Flow	2nd (60–80)	2nd (70–75)	3rd (20–25)	X
Ni-Cd	4th (40–60)	4th (60–80)	4th (5–10)	X
Lead Acid	5th (30–50)	5th (60–70)	5th (3–6)	X

Li-ion = lithium-ion, Na-S = sodium-sulfur, Ni-Cd = nickel-cadmium.

Source: Korea Battery Industry Association 2017 "Energy storage system technology and business model".

Duck Curve


Maximum active power
Reference of active power
P(W)
Maximum power
Power reserves
Power curtailment
Maximum power
Ramp down
Ramp up
t(h)



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BATTERY TECHNOLOGIES

Vented Lead–Acid Battery

- Oldest battery type, in commercial use for over 100 years.
- Operate from the action of lead and lead oxide electrodes, typically with sulphuric acid electrolyte.
- **Advantages**
  - Can deliver high currents.
  - Easily recycled through existing channels.
- **Disadvantages**
  - Uses lead, which is harmful to the environment in disposal and potentially harmful to health for maintainers.
  - Not suited to complete discharge, which may irreversibly damage the battery.
  - Charges slowly.
  - Releases hydrogen gas when charging, so fresh air ventilation is required. Electrolyte maintenance is required. The acid electrolyte is caustic.

Valve-Regulated Lead–Acid Battery

- Sometimes known as 'sealed' batteries because they require no maintenance.
- Principally, there are two varieties: 'acid-gel' electrolyte or absorbent glass mat. Both operate on a similar electrochemical principal to VLA.

Valve-Regulated Lead–Acid Battery

- **Advantages**
  - No maintenance required.
  - Can deliver high currents.
  - 'Deep cycle' versions are available that permit more complete discharge than VLA.
  - Less susceptible to electrolyte leakage if broken.
  - Easily recycled through existing channels.
- **Disadvantages**
  - Uses lead, which is harmful to the environment in disposal.
  - Overcharging can damage the battery.
  - In overcharge or fault conditions, certain VRLA batteries may produce hydrogen, and fresh air ventilation is therefore required.

Lithium-Ion (Li-Ion) Battery

- A more recent technology, these batteries charge and discharge by the movement of lithium ions between the electrodes.
- Originally found popularity in the portable consumer and business electronic products market; advances in product safety and overall performance are now being driven by their use in the automotive electric vehicle market.
- Li-Ion batteries need protection systems to prevent thermal runaway.

Lithium-Ion (Li-Ion) Battery

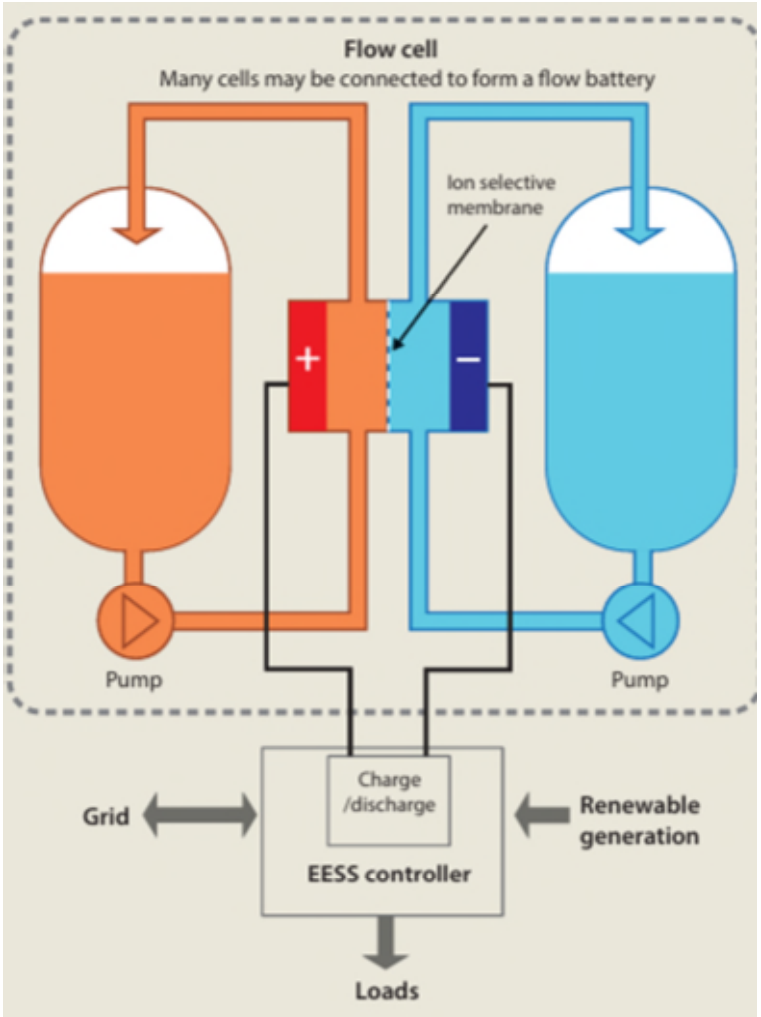
- **Advantages**
  - Lighter, having between 2 and 5 times better storage capacity in terms of weight of the battery than lead-acid types.
  - Between 10 and 20 times more chargedischarge cycles than lead-acid types, making them more suitable for frequent charge-discharge uses associated with loadprofile shifting and domestic generation technologies such as solar PV.
  - Operational life similar to VRLA.

Lithium-Ion (Li-Ion) Battery

- **Disadvantages**
  - Without protection systems, can be subject to thermal runaway.
  - Not easily recyclable at present.
  - Mechanical damage may lead to thermal runaway if not adequately protected against.

Flow Battery

- A very recent technology, where the battery operates somewhat similarly to a reversible fuel cell, or a large battery where the electrolyte is stored in tanks and pumped past electrodes, to increase storage capacity without increasing electrode dimensions.



Flow Battery

- **Advantages**
  - Increasing storage capacity does not mean a corresponding increase in short-circuit current.
  - Thermal conditions in the cell are better managed as flowing electrolytes remove heat from the cell.
  - Potentially longer service life.
  - Can be tailored for very large EESS.

Flow Battery

- **Disadvantages**
  - Requires energy to drive pumps. This is potentially far less significant for larger EESS, but may present inefficiencies for smaller EESS. *Note: Flow batteries therefore may be better suited to community EESS than individual dwellings.*
  - Substances used in certain flow batteries, such as vanadium, may be harmful to the environment in disposal.



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QUESTIONS?